

Question number	Answer	Mark
4(a)	The only correct answer is D. B is incorrect because all three statements are true C is incorrect because all three statements are true D is incorrect because all three statements are true	(1)

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that includes the following points: - attaches to the {malto porin / <i>E. coli</i> / bacteria} (1) - so that the {DNA / genetic material / core} can penetrate into the (bacterial) cell (1)	IGNORE host cell DO NOT ACCEPT RNA IGNORE ref to virus entering cell	(2)

Question number	Answer	Additional guidance	Mark
4(c)(i)	to respire / to make ATP / as an energy source (1)	ACCEPT regulation of gene expression formation of peptidoglycan DO NOT ACCEPT to make energy NB there may be other correct answers that will need checking	(1)

number			
4(c)(ii)	Points relating to the bacterial cell: • mutation in the {lamB gene / gene coding for maltoprotein} • changing the structure of the maltoprotein • so that the J protein can no longer {bind / bind as well} • mutation resulting in the production of a (new) enzyme • that can cleave the J protein off the maltoprotein • preventing E. coli from being {infected / destroyed} • bacteria that survive will divide (asexually) • forming a clone of resistant bacteria • presence of virus acting as a selection pressure • but the maltoprotein would need to remain functional • otherwise no sugars for respiration Points relating to the λ phage: • without being able to attach to the bacteria, the virus will not persist • as it needs a host cell to replicate • being able to bind is the selection pressure • mutation in the viral DNA coding for the J protein • that would result in a J protein capable of binding to the altered maltoprotein • that would result in a J protein that could bind more strongly to the maltoprotein • that would result in binding to another attachment site • viruses that can bind will result in production of viruses that can also	Level 1: 1 mark = 1 relevant comment made 2 marks = 3 points made Level 2: 3 marks = 4 points made that include both virus and bacteria 4 marks = 5 points made that include both virus and bacteria Level 3: 5 marks = 6 points made that include both virus and bacteria 6 marks = 6 points which include Either the need for the maltoprotein to remain functional Or how mutations are passed on to next generation of viruses and bacteria NB limited to max 3 marks if no reference made to either J protein or maltoproteins	(6)

	bind • as their genetic material will be cloned inside the host cell		
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